

OLIVIA THOMMANA

oliviathommana@gmail.com | 9895467609 | Thrissur, Kerala | [linkedin.com/in/olivia-thommana](https://www.linkedin.com/in/olivia-thommana) | github.com/oliviathommana

PROFESSIONAL SUMMARY

Motivated and detail-oriented Computer Science and Engineering student (CGPA 7.88/10) at Christ College of Engineering (Autonomous), Irinjalakuda, with a strong track record of converting academic knowledge into working, real-world systems. Proficient in Python, Java, Embedded C, and web technologies; experienced in designing and deploying machine learning pipelines, computer vision systems, embedded hardware solutions, and database-backed desktop applications. Actively engaged in national hackathons, open-source communities, Google Cloud innovation events, and IEEE activities — combining competitive problem-solving with collaborative team engineering. Eager to contribute to challenging, impact-driven roles in software development, data science, or emerging technology.

EDUCATION

B.Tech – Computer Science and Engineering 2023 – 2027 (Pursuing)

Christ College of Engineering (Autonomous), Irinjalakuda | APJ Abdul Kalam Technological University
Result: CGPA 7.88 / 10 — Consistent academic performance; core subjects include DSA, OS, DBMS, CN, ML, and Software Engineering

Higher Secondary Certificate – Computer Science Stream 2022 – 2023

Don Bosco HSS, Irinjalakuda | Kerala Higher Secondary Board
Result: 98% in Computer Science — Ranked among top performers in the batch; strong foundation in programming logic and mathematics

Secondary School Leaving Certificate (SSLC) 2020 – 2021

Christ Vidyanikethan, Irinjalakuda | Kerala SSLC Board
Result: 94% — Distinction in Mathematics, Science, and IT; built early interest in computing and problem-solving

TECHNICAL SKILLS

Programming: Python, C, Java, Embedded C	Web: HTML5, CSS3 (responsive layouts, semantic markup)
ML & Data: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn	Database: MySQL (DDL/DML, joins, stored procedures, JDBC)
ML Concepts: Regression, Classification, Clustering, Feature Engineering	Soft Skills: Technical writing, team collaboration, public presentation

INDUSTRY INTERNSHIP

Machine Learning Intern · Industry Training Program June 2024 · One Week

Completed an intensive, structured ML internship as part of a 5-member team, covering the full end-to-end supervised and unsupervised learning workflow on real-world datasets under professional supervision.

- Designed and executed a complete ML pipeline: raw data ingestion, null-value treatment, outlier detection, feature scaling (StandardScaler/MinMaxScaler), feature selection (correlation matrices, SelectKBest), model training, cross-validation, and held-out test evaluation.
- Implemented and benchmarked four supervised algorithms — Linear Regression (continuous prediction, MSE/R² metrics), Logistic Regression (binary classification, ROC curves), Decision Trees (interpretable rule generation, Gini/entropy split criteria), and K-Nearest Neighbours (distance-weighted voting, optimal K via elbow method).
- Applied K-Means Clustering for unsupervised patient-group segmentation; determined optimal cluster count using elbow plots and silhouette analysis; visualised clusters with PCA-reduced 2D scatter plots.

PROJECTS

PARQ – Smart Parking Management System

A real-time, camera-based parking occupancy detection system automating slot monitoring and delivering live availability updates on low-cost edge hardware.

- ▶ Built a multi-stage CV pipeline: MJPEG stream → perspective warp → Gaussian blur → adaptive thresholding → morphological filtering → contour extraction → per-slot binary classification with EMA smoothing to suppress false triggers.
- ▶ Deployed on Raspberry Pi 4B with Camera Module V2; offloaded display rendering to a secondary thread, maintaining 15 fps inference at sub-200 ms end-to-end latency.

Tech Stack: Python 3.10, OpenCV 4.8, NumPy, Picamera2, Raspberry Pi 4B, Camera Module V2, JSON config, HTTP server

Automated Egg Incubator – YIP Submission

A microcontroller-driven egg incubator maintaining precise temperature and humidity across the 21-day cycle — submitted to Kerala's Young Innovators Programme (YIP) as a low-cost agri-tech solution for small-scale poultry farmers.

- ▶ Programmed Arduino Uno in Embedded C with a 5-second DHT22 sampling loop; two-stage hysteresis controller manages heating coil and ultrasonic misting relays with inner dead-band to prevent chatter and outer band for emergency alarm.
- ▶ Wired full control circuit on custom stripboard: DHT22 with 10 kΩ pull-up, two optoisolated 5V relay modules driving 230V AC loads, 16×2 I²C LCD for live status display, and piezo buzzer with distinct tonal codes for temperature and humidity faults.

Tech Stack: Embedded C, Arduino Uno, DHT22, 16×2 I²C LCD, Dual Relay Module, Servo Motor, LM393 Comparator, MOSFET, SLA Battery

Holographic Display System

A multi-angle 3D holographic visualisation system using Pepper's Ghost optics, combining original 3D asset creation in Blender, a custom Python compositing pipeline, and precision acrylic hardware construction.

- ▶ Modelled six 3D objects in Blender using polygon modelling and subdivision surfaces; rigged 30 fps keyframe animations with emission-based materials on transparent backgrounds for maximum holographic contrast.
- ▶ Wrote a Python/OpenCV compositing script that applies per-quadrant affine transforms (0°/90°/180°/270° + mirror) onto a 1080×1080 canvas in the cross-pattern layout for a four-sided pyramid reflector, re-encoded to MP4 with H.264.

Tech Stack: Python 3, OpenCV 4, NumPy, Blender 3.6, Tkinter, FFmpeg, Acrylic Pyramid Reflector, 1080p Display

CredifAI – Fake News Detection System · Hackathon, VIT Chennai

An NLP-powered misinformation detection system with a React/Tailwind CSS frontend and Node.js backend, presented at a national-level hackathon among 200+ competing teams.

- ▶ Built the full-stack application with React and Tailwind CSS on the frontend and Node.js on the backend, using Axios for API communication between layers.
- ▶ Integrated an ML classifier to flag fake news in real time; selected for live demonstration to industry judges at VIT Chennai.

Tech Stack: React, Tailwind CSS, Node.js, Axios

Trash2Cash – AI-Based Waste Management Solution · National Hackathon

An AI-based waste classification web app enabling users to identify waste categories and earn recycling incentives; selected for a national-level hackathon modeled on Singapore's urban waste management system.

- ▶ Designed the React frontend and MongoDB backend, integrating an intelligent recommendation engine to classify waste and suggest appropriate recycling actions.
- ▶ Selected for presentation at a national-level hackathon; problem statement modeled on Singapore's urban waste infrastructure, addressing scalability and incentive-based civic engagement.

Tech Stack: React, MongoDB

CareFlow – AI Intelligent Caregiver · Hackathon, RSET Kochi

An AI-driven caregiver workload optimization engine for elderly care, automating intelligent task scheduling and assignment based on real-time patient needs.

- Developed caregiver assignment logic using the Groq API to power intelligent scheduling, balancing workloads dynamically against real-time patient care requirements.
- Built the React frontend and FastAPI backend, integrating the Groq API for the intelligent scheduling engine presented at RSET Kochi hackathon.

Tech Stack: *React, FastAPI, Groq API*

CERTIFICATIONS

NPTEL (IIT/IISc) — Computer Architecture — CPU design, pipelining, cache hierarchy, memory systems, instruction sets 2024

Coursera (Google) — Foundations: Data, Data Everywhere — data analytics lifecycle, spreadsheets, SQL queries, data-driven decision-making 2024

Coursera — Use Canva to Design Digital Course Collateral — visual design, typography, layout principles, digital content production 2024

ACTIVITIES, HACKATHONS & WORKSHOPS

- Adobe India Hackathon (Online) — cleared competitive coding assessment round; solved algorithmic and data-structure problems under timed conditions alongside participants from premier engineering colleges across India.
- Hackathon at VIT Chennai — contributed to a team-based 24-hour build challenge; led ideation and technical prototyping, culminating in a live product demonstration to industry judges.
- Google Cloud Agentic AI Day — submitted a conceptual proposal for an AI-powered agentic system; recognised for innovative application of LLM orchestration and Google Cloud infrastructure.
- Abaja National Level Technical Competition — represented institution nationally; demonstrated algorithmic problem-solving and cross-disciplinary technical knowledge in judged rounds.
- Workshops attended: Artificial Intelligence & ML, Cybersecurity & Ethical Hacking (OWASP Top 10, pen-testing basics), Web Development (HTML/CSS/JS), Ultimate Game Design, and Dhrishti Project Expo (project presentation to industry panel).
- Industrial Visit — Spectrum Softtech Solutions (P) Ltd — observed professional agile workflows, sprint planning, QA processes, and end-to-end client delivery pipelines in a live commercial software environment.

ORGANISATIONS & COMMUNITY

IEDC · Member

- Attended IEDC Summit by Kerala Startup Mission — engaged with sessions on product ideation, design thinking, lean startup methodology, and funding pathways; applied insights directly to personal project development.

FOSS Club & IEEE SB · Member — Relay Coding (FOSS) & Brain Battle (IEEE)

- Competed in Relay Coding: a timed team event where each member programs for a fixed interval before handing off, testing both individual skill and seamless team communication under pressure.
- Participated in Brain Battle (IEEE SB): a multi-round event combining logical reasoning, technical MCQs, and live coding puzzles designed to challenge CS breadth and analytical depth.

NSS · Member

- Contributed regularly to community service campaigns — campus cleanliness, blood donation awareness, and outreach to local schools — building social responsibility and leadership alongside academic commitments.

LANGUAGES

English — Professional working proficiency (reading, writing, speaking, technical documentation) **Malayalam** — Native / full proficiency **Hindi** — Conversational working knowledge